



JAIDEV EDUCATION SOCIETY'S
J D COLLEGE OF ENGINEERING AND MANAGEMENT
KATOL ROAD, NAGPUR

Affiliated to Dr. Babasaheb Ambedkar Technological University, Lonere

Website: www.jdcoem.ac.in E-mail: info@jdcoem.ac.in

An Autonomous Institute, with NAAC "A" Grade
Department of Computer Science and Engineering

"A Place to Learn, A Chance to Grow"

Session-2025-26



VISION

To be recognized as an Excellent Innovative Engineering Department through Academic Programme which develops leaders in Education and Research in Computer Science

MISSION

1. To create self-learning environment by facilitating leadership qualities, team-spirit and ethical responsibilities.
2. To strengthen department-industry collaboration and interaction with professional society through technical knowledge and internship program.
3. To promote research and development with current techniques through well qualified resources in the area of computer science and wireless engineering.

Question Bank

Branch: - Computer Science and Engineering

Semester / Year: - V Sem / 3rd Year

Subject: - Interactive Graphics

Subject Code: - CS5E001C

UNIT 1: Display Devices

✓ 2 Marks

1. Define raster and vector displays.
2. What is a digital frame buffer?
3. List any two colour display techniques.
4. Define pixel and point plotters.
5. What is meant by character generator?

✓ 4 Marks (Understanding/Applying)

1. Differentiate between raster and vector scan systems with diagrams.
2. Explain the working of a plasma panel display.
3. Describe any two high-speed drawing devices.
4. Explain colour look-up tables and shadow mask CRT with diagrams.

✓ 7 Marks (Applying/Analysing)

1. Compare continual refresh and storage displays. Mention their pros and cons.
2. Explain the architecture and functions of a display processor.
3. Discuss different colour display techniques and their use cases.

✓ 12 Marks (Evaluating/Creating)

1. Evaluate the importance of various display devices in modern graphics systems with suitable examples.



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2. Explain in detail the complete working of a character generator and high-resolution devices with neat sketches.

UNIT 2: Display Description

✓ 2 Marks

1. Define screen coordinates and user coordinates.
2. What is a vector list?
3. What are homogeneous coordinates?
4. State the use of compressed incremental list.

✓ 4 Marks

1. Explain the difference between screen and user coordinates.
2. Describe any two graphical data structures.
3. What is the view algorithm? Explain briefly.
4. List and explain any two display code generation techniques.

✓ 7 Marks

1. Apply the concept of two-dimensional transformation with suitable example.
2. Describe line and circle drawing algorithms used in computer graphics.

✓ 12 Marks

1. Design and explain a system using graphical data structures and view algorithm for a 2D interface.
2. Evaluate different transformation techniques and their role in display systems.



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UNIT 3: Interactive Graphics

✓ 2 Marks

1. Define light pen and digitizing tablet.
2. What is meant by interactive graphical technique?
3. List any two pointing and positioning devices.
4. Define panning and zooming.
5. What is meant by clipping in interactive graphics?

✓ 4 Marks

1. Differentiate between mouse and trackball.
2. Explain rubber-band line technique with example.
3. Describe the use of linking in interactive graphics.
4. Write short notes on windowing and scissoring.

✓ 7 Marks

1. Illustrate various interactive positioning techniques with neat diagrams.
2. Explain clipping operations for interactive graphics with examples.
3. Describe the role of mouse programming in interactive graphics.

✓ 12 Marks

1. Analyse interactive graphical techniques such as zooming, panning, clipping, and scissoring with suitable examples.
2. Design an interactive system using mouse and light pen for a CAD application.



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UNIT 4: 2D Geometric Transformations & Viewing

✓ 2 Marks

1. Define reflection and shearing transformation.
2. What is a composite transformation?
3. State the purpose of homogeneous coordinates.
4. Define viewing pipeline in 2D graphics.
5. What is line clipping?

✓ 4 Marks

1. Explain matrix representation of translation, scaling, and rotation.
2. Differentiate between window and viewport.
3. Describe point clipping and polygon clipping.
4. Write short notes on text clipping.

✓ 7 Marks

1. Apply composite transformations on a given 2D object with an example.
2. Illustrate Cohen-Sutherland line clipping algorithm with example.
3. Explain the complete window-to-viewport coordinate transformation.

✓ 12 Marks

1. Evaluate different 2D geometric transformation techniques with real-time applications.
2. Design a clipping pipeline for polygon, line, and text clipping operations.



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UNIT 5: 3D Graphics

✓ 2 Marks

1. Define wireframe model.
2. What is perspective depth?
3. Define hidden line and hidden surface removal.
4. State the purpose of shading in 3D graphics.
5. What is a projective transformation?

✓ 4 Marks

1. Differentiate between parallel and perspective projections.
2. Explain transparent solids with an example.
3. Describe two types of shading techniques.
4. Write short notes on GUI in 3D graphics.

✓ 7 Marks

1. Explain perspective transformation and its applications in 3D graphics.
2. Illustrate hidden surface elimination algorithms with suitable examples.
3. Apply 3D transformations (translation, scaling, rotation) on a cube.

✓ 12 Marks

1. Evaluate wireframe, hidden line, surface elimination, and shading techniques in 3D graphics with applications.
2. Design a 3D graphics pipeline showing transformations, projections, and rendering with neat diagrams.